State of the Ecosystem 2026: Request Tracking Memo

Introduction

In this document, we summarize a running list of comments and requests from the Mid Atlantic Fishery Management council (MAFMC), New England Fishery Management Council (NEFMC), and both Science and Statistical Committees (SSCs). The memo is organized by task based on its current status and topic category. The Priority column (formerly called Rank) has been updated based on quantitative scoring guidelines that were developed following the publication of 2025 State of the Ecosystem (SOE) reports. Priority determination methods and scores were discussed with Council staff and SSC members at the July 2025 State of the Ecosystem Prioritization meeting. We first show requests that have been completed as of the 2026 SOE reports (Table 1), research requests that are in progress or not yet started (Table 2), and requests that have been removed from tracking due to being out of scope or not well-defined (Table 3).

Table 1: Requests that have been completed as of the 2026 State of the Ecosytem reports.

Request	Year	Priority
SOE admin		
Include estimates of inclusion years in request memo	2022	Lowest
System level thresholds/ ref pts		
Trend Analysis	2019 - 2024	Highest
Multiple system drivers		
Profits vs Revenue: net revenue indicator incomplete and different trend from gross	2023-2024	Highest
Clarify objectives, terminology and presentation for fishing community engagement/reliance/social indicators	2022, 2024	Highest
Time series of social indicators	2023-2024	Highest
Consider appropriate scale for social and economic indicators	2024	Highest
Add social and economic considerations in the Climate and Ecosystem risks section	2024	Highest
Clarify community definitions and consider indicators beyond landings to employment, subsistence	2024	Highest
Include community affordability in port level vulnerability	2024	High
Report changes in small fish and large fish biomass along with production anomalies	2024	Moderate
Relate OA to nutrient input; are there "dead zones" (hypoxia)?	2021	Low
Estuarine Water Quality	2020	Low
What determines a "risk"? Include aquaculture as a risk?	2022	Unranked

Table 2: Remaining research requests for the State of the Ecosystem reports.

Request	Year	Status	Priority
Short term forecasts			
Short term forecasting from CEFI (water temp, productivity); Characterize current conditions in context of expected short term change	2022, 2024, 2025	In progress	Highest
Using phytoplankton trends to forecast fish stocks	2022	Not started	Moderate
Management			
Ensure that indicators are formatted in a way that is useable for risk work	2024	Not started	High
Management complexity	2019	In progress	Moderate
Recreational bycatch mortality as an indicator of regulatory waste	2021	Not started	Moderate
Functional group level status/ thresholds/ ref pts			
Biomass+landings trends by (aggregated) Council managed species, stock status, cold vs warm water spp	2024	In progress	High
VAST	2020	In progress	Moderate
Expand HMS and rec indicators to include tunas and commercially caught HMS species to explore further evidence of distribution shifts	2025	In progress	Moderate
Consider expanding species included in seabird productivity indicators–Roseate tern	2024	Not started	Low
Seal index, Apex predator index (pinnipeds, time series of marine mammal species abundance	2020, 2021, 2024	In progress	Low
Stock level indicators			
Include cross references to stock specific products (ECSA/ESP), ensure consistent approaches; Establish more links between events and consequences (e.g. temp ranges for more species)	2024	In progress	Highest
Shellfish growth/distribution linked to climate (system productivity)	2019	In progress	Moderate
Indicator of scallop pred pops poorly sampled by bottom trawls	2021	Not started	Moderate
Update text around NARW trend drivers, specifically recent stabilization (for PSD)	2024	Not started	Moderate
Evaluate if NARW hotspots changing over time	2024	Not started	Low
SOE admin			
Present relevant MA information in NE report	2025	In progress	Highest
Improve context and definitions of Management Objectives	2025	Not started	Highest
Include statement on reliance on contributors for evaluation of data quality	2024	Not started	High
Establish criteria for inclusion of events in the Year Highlights section	2024	In progress	High
Present the SOE highlights page to Advisory Panels for feedback	2025	Not started	Moderate
SOE usage tracking	2022-2023	In progress	Low
System level thresholds/ ref pts			
Ecosystem Overfishing Indicators: compare to empirical thresholds; assess informativeness of indicators using simulation analysis; assess impact of phytoplankton size composition on EOF thresholds; determine optimum yield	2021-2024	In progress	Highest
Develop regime shift analysis methods (e.g., inflection point analysis, influence statistics, break point analysis, early warning variance)	2019-2025	Not started	Highest
Conduct regime shift analysis of relevant indicators (e.g., zooplankton, forage fish, socioeconomic, etc). Use influence statistics to identify whether we are approaching tipping points.	2019-2025	Not started	Highest
Use community and port level information to inform fleet stability indicator	2025	In progress	Highest
Update CVA and explore trend changes when a new CVA is applied	2025	In progress	Highest
Include standardized language about uncertainty from e.g. IPCC or NCA applicable to each indicator or data input	2024	Not started	Highest
Create volatility indices to assess system stability	2025	Not started	High
Improve language around status and trends (e.g., assumed baselines, contextualize long v short trends, contextualize local influences)	2025	In progress	High
Sum of TAC/ Landings relative to TAC	2021, 2024	In progress	High
Present normalized indicators to more clearly show variability; incorporate into assessment of system stability	2025	Not started	Moderate

Table 2: Remaining research requests for the State of the Ecosystem reports.

Request	Year	Status	Priority
Explore stock assessment indices as measure of stability	2025	Not started	Moderate
Better evaluation of one-time observations (i.e. address persistence of observations)	2025	Not started	Low
Nutrient input, Benthic Flux and POC (particulate organic carbon) to inform benthic productivity by something other than surface indicators	2021, 2023	Not started	Low
Multiple system drivers			
Evaluate port level landings considering home port vs out of state vessel landings	2024	Not started	High
Consider including indicators of predator prey relationships / predator pressure	2024	In progress	High
Young of Year index from multiple surveys	2019	Not started	High
Tell Social stories like we try to tell biological stories	2022	Not started	High
Vessel-level diversity vs fleet level diversity. Recontextualize fishery stability as adaptive capacity/flexibility.	2023	In progress	High
Distribution shifts: percent of species shifting in similar directions and magnitude of the change, id cross jurisdictional shifts	2024	In progress	High
Relate declining commercial landings to interactions with Council and non-Council species. Especially focus on larger vessels with multiple permits that can shift fishing targets from year to year	2025	Not started	High
Assess if species distributions and processing capacity are drivers of profitibility	2025	Not started	High
Linking Condition	2020	In progress	Moderate
New diet indicators: average weight of diet components by feeding group, mean stomach weight across feeding guilds	2019	Not started	Moderate
Cumulative weather index	2020	In progress	Moderate
Modeling cold pool/warm core ring and wind development interactions	2022	Not started	Moderate
Impact of climate on data streams (changes in catchability of survey)	2022	In progress	Moderate
OA linked to scallop harvest in areas where aragonite saturation is highlighted.	2023	Not started	Moderate
Stability indicator - yield over time in NE	2023	Not started	Moderate
Inclusion of upcoming HMS climate vulnerability assessment	2023	Not started	Moderate
Aggregate surplus production and avg length of inds at recruitment age to productivity section	2024	Not started	Moderate
Consider HMS recreational effort in effort trends	2024	In progress	Moderate
Changing per capita seafood consumption as driver of revenue?	2021	Not started	Moderate
Decomposition of diversity drivers highlighting social components	2021	Not started	Moderate
Assess connection between community vulnerability indicators and recreational fleet diversity and shore-based fishing mode	2025	Not started	Moderate
Assess if there is generational shift in fishing activity; test if this is linked to shore-based fishing	2025	In progress	Moderate
Capture risks to climate for shore anglers in cultural objectives/risk metrics	2025	Not started	Moderate
Qualify the linkage between slope water and cold pool persistence	2025	Not started	Moderate
Investigate the role of spatial distribution of costs and consolidation of industry into fewer ports	2025	Not started	Moderate
Explore the impact of changed timing on reproductive success, climate change vulnerability and recruits	2025	Not started	Moderate
Links between species availability inshore/offshore (estuarine conditions) and trends in recreational fishing effort?	2021	In progress	Low
Track subsequent impacts of 2023 phytoplankton bloom in GOM. Track anomalous events from previous years.	2024	In progress	Low
Consider and account for the role of labor in cost calculations	2025	Not started	Low

Table 3: Requests that have been removed from tracking due to being out of scope or not well-defined.

Request	Year
System level thresholds/ ref pts	
Reduce indicator dimensionality with multivariate statistics	2020
Management	
Re-evaluate EPUs	2020
conduct a tradeoff analysis to determine which management objectives have the largest impact and should be prioritized for monitoring, similar to the summer flounder MSE.	2025
Identify technical interactions within fisheries so management does not cause underfishing	2025
Multiple system drivers	
Inclusion of additional environmental justice indicators from National Academies	2024
Consider state managed fishery production instead of recreational shark harvest	2024
Consider HMS predation pressure	2024
Indicators of chemical pollution in offshore waters	2021
Estuarine condition relative to power plants and temp	2019
Functional group level status/ thresholds/ ref pts	
Uncertainty	2020
Biomass of spp not included in BTS	2020
Stock level indicators	
Sturgeon Bycatch	2021
SOE admin	
Include estimates of inclusion years in request memo	2022
More frequent indicator updates in web based product	2024

Quantitative priority rating criteria

In order to use time more effectively, we developed criteria to guide and speed up the prioritization of requests. These criteria are framed around the impact of the request on the scientific content of the SOE, combined with the impact on fisheries management decision making, regardless of currently available resources. Improvements to the scientific content of the SOE were considered on the ecosystem scale, not on the scale of how that information would affect a single species. Impact on management decision making was characterized as how well the information would fit into a well-defined on-ramp into the management process, such as a risk assessment. Requests that had been previously ranked as "lowest priority" were re-assessed according to the new criteria, and were removed from the tracking list if the request was out of scope or not well-defined.

The new priority definitions are:

- Highest: Significant scientific improvement to the SOE **and** maximum impact on management decision making
- High: Significant scientific improvement to SOE **or** maximum impact on management decision making
- Moderate: The impact on management decision making is unclear, but potentially impactful
- Low: Negligible scientific improvement to SOE and minimal impact on management decision making

Please note that some requests are currently marked as "low priority" due to the imprecise or vague wording, which could be updated in the future if the request is made more specific. All rankings can continue to be discussed and refined in the next SOE Prioritization meeting.

Overview of requests

Request categories

This memo is organized into categories by topic. The former category called "Regime shifts" has been absorbed into the "System level thresholds/reference points" category. The ability to work on requests subject to resource limitations and staffing. The request categories are:

Short term forecasts Includes requests for biological and environmental forecasts, including MOM6 oceanographic model products.

Management Includes analyses related to management performance.

Stock level indicators Includes requests for information on the scale of a single species or stock. Requests for this information may be more appropriately directed to stock specific ecosystem products such as Ecosystem and Socioeconomic Profiles (ESPs).

Functional group level status/thresholds/reference points Includes requests for information on the scale of functional groups or guilds.

SOE admin Includes requests relating to the structure and methods of the creation of the SOE reports.

System level thresholds/reference points Includes requests to develop analytical methods that can be applied across all indicator types and operationalized for management advice.

Multiple system drivers Includes requests for information about oceanographic, ecological, social, and economic processes that impact fisheries.

We welcome further feedback on planned and continuing work.

Adjustments to general SOE report structure

Based on feedback and requests for information surrounding our Stability section, we revised both the ecosystem and fishery stability sections to focus on Changes From Baseline, Adaptive Capacity, and Volatility. This reframing is intended to better communicate how the system has changed from historic observations, how well-prepared the system is for future change, and how variable the system is.

We have retained and updated the online indicator catalog that provides a "deep dive" into each indicator, with multiple visualizations of the data and clearer links to the datasets in the ecodata R package for increased transparency and ease of use by investigators throughout the region. We are also updating the online technical documentation.

Summary of completed requests by category

SOE admin

We have reformatted the memo to identify which requests have been completed in the last year, as well as which requests have been dropped from future tracking.

System level thresholds/reference points

A refined trend analysis is now systematically incorporated into the SOE and the ecodata R package. Both long-term and short-term trends are assessed and plotted if statistically significant at the $p < 0.05$ level. To minimize bias introduced by small sample size, time series with less than 30 years are assessed using the short-term trend methodology applied over the extent of the time period.

OA and hypoxia indicators have been developed through collaboration with the FishBot team. Estuarine water quality is described in the Chesapeake Bay Water Quality Standards Attainment catalog page.

Multiple system drivers

The profitability indicator developed by Geret DePiper (TAMUCC) has been incorporated into the report for the second year in a row, and this request has been removed from tracking.

Experts from the Social Sciences Branch contributed edits, information, and updated methods to the fishing community social indicators. Part of this effort included the development of time series of fishing activity by port. We have also begun including Committees at Sea indicators for both regions to address alternative community definitions. Although methods will continue to be updated, we consider the basic requirements of the social indicator requests to have been met and remove them from tracking.

Aquaculture has been added to the Other Ocean Uses section of the reports. We will continue to expand this section in future reports.

Productivity anomaly indicators have been included for small fish to large fish biomass and recruitment anomalies from stock assessments.

Requests in Progress

Of the 90 requests we are actively tracking, 28 of them are currently *in progress*. Instead of providing a detailed summary of the status of each request, below is a high level summary of the ongoing activities that we are aware of. The following activities represent active work that relates to specific requests, but is not comprehensive of all research being done elsewhere.

SOE admin

The SOE Team has been expanding the reach of their SOE Highlights section and is planning on revising the inclusion criteria following a workshop during the 2026 Cooperative Research Summit.

System level thresholds/reference points

The revision of the Ecosystem and Fishery Stability section has led to discussions about how to add or modify existing indicators to better suit this definition. Research on ecosystem overfishing is also being conducted with ecosystem models that may lead to revisions to prior ecosystem overfishing indicators and the establishment of ecosystem thresholds in the SOE. As work concludes on the Climate Vulnerability Assessment (CVA) 2.0, the SOE Team is considering how to develop related indicators.

Multiple system drivers

There is ongoing research at the NEFSC on fishery adaptation strategies, demographic shifts, and shifts in fishing behavior that may lead to future indicators. Species distribution models from the CVA2.0 may address requests to quantify species distribution shifts. Work continues to evaluate changes in social and ecosystem changes relating to highly migratory and protected species.

Short term forecasts

An operational Northwest Atlantic MOM6 forecast model has been developed which provides quarterly 12-month forecasts of temperature and other oceanographic variables for the entire Northeast U.S. A high-level infographic has been included into the SOE summary page presenting the 2026 bottom temperature forecasts and the 10 year projection. We plan to continue integrating forecast results into the SOE when appropriate.